



SUSTAINABLE OLYMPIAD

Sustainable Olympiad 2025

Round 1: Online Knowledge Challenge

Senior Division (ages 15–18)

Time allowed	90 minutes
Total marks	60
Part A	10 multiple-choice questions, 20 marks
Part B	5 structured problems, 40 marks

Instructions

- Answer **all** questions. Work together as a team.
- Part A: choose **one** answer (A, B, C or D) for each question. There are no negative marks.
- Part B: show all your working. Marks are given for correct method even if the final answer is wrong.
- Give numerical answers to an appropriate number of significant figures, with units.
- Scientific calculators are allowed. Internet access and AI tools are not allowed during the round.
- A data sheet is provided on the next page.

Team name

School

Data sheet

- Acceleration due to gravity: $g = 9.81 \text{ m s}^{-2}$
- Specific heat capacity of water: $c = 4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$; density of water 1.00 kg L^{-1}
- $1 \text{ kWh} = 3.6 \text{ MJ}$; $0 \text{ }^{\circ}\text{C} = 273.15 \text{ K}$; $1 \text{ year} = 8760 \text{ h}$
- Planck constant $h = 6.63 \times 10^{-34} \text{ J s}$; speed of light $c = 3.00 \times 10^8 \text{ m s}^{-1}$; $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
- Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
- Molar masses (g mol^{-1}): H 1.008, C 12.01, N 14.01, O 16.00
- Radiative forcing of CO_2 : $\Delta F = 5.35 \ln(C/C_0) \text{ W m}^{-2}$

Part A: Multiple choice (20 marks)

Each question is worth 2 marks. Choose one answer.

A1. A rooftop solar module has an area of 1.6 m^2 and an efficiency of 20%. On a day equivalent to 4.5 peak sun hours at 1000 W m^{-2} , approximately how much electrical energy does it produce? **[2]**

- A** 0.32 kWh
- B** 1.44 kWh
- C** 7.2 kWh
- D** 14.4 kWh

A2. Over 20 years, the global warming potential (GWP) of methane is about 81, but over 100 years it is only about 28. What is the main reason for this difference? **[2]**

- A** Methane absorbs infrared radiation more strongly as its concentration rises
- B** Methane has an atmospheric lifetime of only about 12 years, so most of its warming happens in the first decades
- C** Methane is oxidised to CO_2 , which has a higher GWP than methane
- D** The 100-year GWP includes absorption of methane by the oceans

A3. According to Betz's law, what is the maximum fraction of the kinetic energy flux of the wind that an ideal turbine can extract? **[2]**

- A** $1/3$
- B** $1/2$
- C** $16/27$
- D** $8/9$

A4. The power available to a wind turbine is $P = \frac{1}{2}\rho A v^3$. If the mean wind speed at a site is 9.0 m s^{-1} instead of 6.0 m s^{-1} , by what factor does the available power increase? **[2]**

- A** 1.5
- B** 2.25
- C** 3.375
- D** 4.5

A5. The average pH of the surface ocean has fallen from about 8.2 to 8.1 since pre-industrial times. By approximately what percentage has the hydrogen-ion concentration increased? [2]

- A 1%
- B 10%
- C 26%
- D 100%

A6. Producers in a grassland fix $50\,000\text{ kJ m}^{-2}\text{ yr}^{-1}$. Assuming 10% of the energy is transferred between successive trophic levels, how much energy is available to tertiary consumers? [2]

- A $5000\text{ kJ m}^{-2}\text{ yr}^{-1}$
- B $500\text{ kJ m}^{-2}\text{ yr}^{-1}$
- C $50\text{ kJ m}^{-2}\text{ yr}^{-1}$
- D $5\text{ kJ m}^{-2}\text{ yr}^{-1}$

A7. In a eutrophic lake, which process is most directly responsible for the sharp fall in dissolved oxygen after an algal bloom? [2]

- A Algae absorbing oxygen through their cell walls during the day
- B Aerobic decomposition of dead algae by bacteria
- C Nitrogen fixation by cyanobacteria
- D An increase in oxygen solubility as the water cools

A8. Which of the following is a **negative** (stabilising) feedback in the climate system? [2]

- A The ice-albedo feedback
- B Release of methane from thawing permafrost
- C The water-vapour feedback
- D Increased emission of infrared radiation from a warmer surface

A9. A geothermal power plant takes heat from a reservoir at $180\text{ }^{\circ}\text{C}$ and rejects heat to cooling water at $30\text{ }^{\circ}\text{C}$. What is its maximum theoretical (Carnot) efficiency? [2]

- A 17%
- B 33%
- C 67%
- D 83%

A10. A heat pump with a coefficient of performance (COP) of 3.5 delivers 14 kW of heat to a building. At what rate does it extract heat from the outside air? [2]

- A 3.5 kW
- B 4.0 kW
- C 10 kW
- D 49 kW

Part B: Structured problems (40 marks)

B1. Decarbonising school heating (10 marks)

A school needs 108 000 kWh of heat per year. The heat is supplied by a gas boiler with an efficiency of 90%. Burning natural gas releases 0.20 kg CO₂ per kWh of gas burned. The school is considering an air-source heat pump with a seasonal COP of 3.0. Grid electricity has an emission factor of 0.25 kg CO₂ per kWh.

(a) Calculate the annual CO₂ emissions from the gas boiler, in tonnes. [2]

(b) Calculate the annual CO₂ emissions if the heat pump supplies all the heat. [2]

(c) Calculate the percentage reduction in emissions. [2]

(d) Above what grid emission factor would the heat pump cause more CO₂ than the boiler? [2]

(e) Suggest two reasons why the real reduction might differ from your answer to (c). [2]

B2. Fuels and combustion (8 marks)

Model petrol as pure octane, C₈H₁₈ (density 0.74 kg L⁻¹, energy released 44.4 MJ kg⁻¹). Ethanol, C₂H₅OH, releases 26.8 MJ kg⁻¹.

(a) Write a balanced equation for the complete combustion of octane. **[2]**

(b) Calculate the mass of CO₂ produced when 1.00 kg of octane burns completely. **[2]**

(c) A car uses 6.0 L of petrol per 100 km. Calculate its CO₂ emissions in g km⁻¹. **[2]**

(d) Calculate the mass of CO₂ released per MJ of energy for octane and for ethanol. Then explain why bioethanol can still reduce net emissions. **[2]**

B3. Estimating and modelling a population (8 marks)

To estimate a population of ground beetles, students capture 60 beetles, mark them harmlessly and release them. A week later they capture 75 beetles, of which 15 are marked.

(a) Use the Lincoln-Petersen index to estimate the population size. **[2]**

(b) State two assumptions of this method. **[2]**

(c) Chapman's less-biased estimator is $N = (M + 1)(C + 1)/(R + 1) - 1$. **[1]**
Calculate this estimate.

- (d)** The population grows logistically: $dN/dt = rN(1 - N/K)$, with $r = 0.40 \text{ yr}^{-1}$ and $K = 500$. Calculate the growth rate when $N = 300$. Then find the population size at which the growth rate is greatest, and that maximum rate. **[3]**

B4. Rainwater harvesting (7 marks)

A school has a roof area of 450 m^2 , and the annual rainfall is 750 mm. On average, 85% of the rain landing on the roof can be collected. The 600 students each flush a toilet twice per school day, using 6.0 L per flush, on 190 school days a year.

- (a)** Calculate the volume of rainwater collected per year, in m^3 . **[2]**

- (b)** What percentage of the school's toilet-flushing water could rainwater supply? **[2]**

- (c)** All the harvested water is pumped up 12 m to a header tank by a pump that is 50% efficient. Calculate the electrical energy used per year, in kWh. **[3]**

B5. Carbon dioxide and global temperature (7 marks)

The pre-industrial CO_2 concentration was $C_0 = 280 \text{ ppm}$. Assume the equilibrium temperature change is $\Delta T = \lambda \Delta F$, with $\lambda = 0.80 \text{ K per W m}^{-2}$. Ignore all other gases and aerosols.

- (a)** Calculate the radiative forcing and the equilibrium warming for a concentration of 420 ppm. **[2]**

(b) Calculate the equilibrium warming for a doubling of CO_2 (the climate sensitivity in this model). **[1]**

(c) At what CO_2 concentration would the equilibrium warming reach 1.5 K? **[2]**

(d) Observed warming is about 1.3 K, even though CO_2 is above your answer to (c). Give two reasons. **[2]**

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